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Effects of oil sands tailings compounds and harsh weather on mortality rates, growth and detoxification efforts in nestling tree swallows (*Tachycineta bicolor*)

Marie-Line Gentes [a,*], Cheryl Waldner [b,1], Zsuzsanna Papp [c,2], Judit E.G. Smits [a,3]

[a] *Department of Veterinary Pathology, University of Saskatchewan, 52 Campus Dr., Saskatoon, SK S7N 5B4, Canada*
[b] *Department of Large Animal Clinical Sciences, University of Saskatchewan, 52 Campus Dr., Saskatoon, SK S7N 5B4, Canada*
[c] *Toxicology Research Centre, 44 Campus Dr., Saskatoon, SK S7N 5B3, Canada*

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Under natural stress caused by harsh weather, birds exposed to chemicals from the oil sands extraction process suffered higher mortality than those in control areas.

Abstract

Oil sands mining companies in Alberta, Canada, are evaluating the feasibility of using wetlands to detoxify oil sands process material (OSPM) as a reclamation strategy. Reproductive success, nestling growth, survival and ethoxyresorufin-*o*-deethylase (EROD) activity were measured in tree swallows (*Tachycineta bicolor*) on experimental wetlands. In 2003, harsh weather triggered a widespread nestling die-off. Mortality rates on the control site reached 48% while they ranged from 59% to 100% on reclaimed wetlands. The odds of dying on the most process-affected sites were more than ten times higher than those on the control site. In 2004, weather was less challenging. Mortality rates were low, but nestlings on reclaimed wetlands weighed less than those on the control site, and had higher EROD activity. These results indicate that compared with reference birds, nestlings from OSPM-impacted wetlands may be less able to withstand additional stressors, which could decrease their chances of survival after fledging.

Keywords: Tree swallows; Oil sands; Mortality; Weather; Contaminants

1. Introduction

The oil sands deposits of north-eastern Alberta, Canada, cover over 140 000 km^2 and contain 15% of the world reserves of crude oil (Alberta Department of Energy, 2005). With the decline of conventional fossil fuels, development of the vast economic potential of the oil sands reserves has become a priority. The separation of bitumen from sand produces large volumes of process-affected water, which are currently held in large settling ponds on the mining sites (Mikula et al., 1996). Operators estimate that as much as 1 billion m^3 of Oil Sands Process-Affected Water (OSPW) could require reclamation when the current mining operations cease (FTFC, 1995). Integration of these large volumes of contaminated water into the environment represents a major challenge for the industry. "Wet landscape" reclamation options include the development of a series of wetlands and Pit Lakes which will contain aged OSPW. Most of the acute toxicity associated with the tailings will be remediated by natural processes such as biodegradation, but some of the OSPW characteristics will remain (FTFC, 1995; Lai et al., 1996). This reclamation strategy is meant to recreate sustainable ecosystems, with wetlands

* Corresponding author. Tel.: +1 306 966 7309; fax: +1 306 966 7439.
E-mail addresses: marie-line.gentes@usask.ca (M.-L. Gentes), cheryl.waldner@usask.ca (C. Waldner), zsp127@duke.usask.ca (Z. Papp), judit.smits@usask.ca (J.E.G. Smits).
[1] Tel.: +1 306 966 7168; fax: +1 306 966 7159.
[2] Tel.: +1 306 966 8731; fax: +1 306 931 1664.
[3] Tel.: +1 306 966 7445; fax: +1 306 966 7439.

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and lakes having an appearance and productivity similar to those present throughout the region (Nix et al., 1993).

Oil sands process-affected water in wetlands has been shown to be toxic to plankton communities (Leung et al., 2003), fish (van den Heuvel et al., 2000) and amphibians (Pollet and Young, 2000). In comparison to most natural surface waters in the oil sands region, the OSPW contains elevated levels of salt, dissolved organics such as naphthenic acids (NAs), as well as polycyclic aromatic hydrocarbons (PAHs). The PAHs result from the presence of unrecovered bitumen in water and sediments. Various studies have shown PAHs to have toxic, mutagenic and carcinogenic properties to fish, birds and laboratory rodents (reviewed by Albers, 2003).

Naphthenic acids (NAs), present in OSPW, are a group of carboxylic acids with surfactant-like properties. When freshly released from the oil sand extraction process, they account for most of the toxicity to aquatic organisms, and have been proven toxic to mammals (Rogers, 2003). With aging and natural bioremediation processes, the acute toxicity is removed, but chronic impacts may still occur (Herman et al., 1994; Rogers, 2003). The exact mechanisms of NA toxicity have yet to be identified.

Biota living on reclaimed wetlands are exposed to OSPW. To assess the sustainability of the wet landscape reclamation strategy before its application on a much larger scale, experimental ponds and wetlands containing OSPW have been constructed during the last 15–20 y by Syncrude Canada Ltd. and Suncor Energy Inc. Tree swallows (*Tachycineta bicolor*) nesting around these wetlands were designated as upper trophic level bioindicators of environmental contamination, because insects of aquatic origin account for 84% of their diet (Smits et al., 2000). Work reported by Ganshorn (2002) indicates that the bioaccumulation potential of PAHs in benthic dipterians inhabiting reclaimed wetlands should be quite low because of Biota-Sediment Accumulation Factors of less than one (0.7). However, considering that concentrations of parent and alkylated PAHs reach relatively high levels on reclaimed wetlands (Smits, 2000), doses delivered to nestlings through their diet may be sufficient to elicit sublethal effects. Data on the bioavailability and bioaccumulation potential of NAs are still lacking because of the difficulties associated with measuring concentrations of this class of chemicals in animal tissues.

Tree swallows are an effective model species, and they have been widely used in environmental impact assessments. In most of these assessments, reproduction has been an important endpoint. A number of studies reported a reduction in reproductive performance on sites affected by chemicals of anthropogenic origin (Bishop et al., 2000a; Custer et al., 2003; McCarty and Secord, 1999), but others did not detect measurable effects (Bishop et al., 1999; Custer et al., 1998; Elliott et al., 1994; Gerrard and St-Louis, 2001; Harris and Elliott, 2000). The biological effects on the swallows were often surprisingly low considering the degree of exposure to contaminants. However, most of these field studies were conducted under favorable meteorological conditions. Birds may respond differently to chemical exposure if they are simultaneously subjected to stressful environmental conditions. The impact of inclement weather on the reproduction of birds nesting on contaminated sites was investigated in only one of these studies (Custer et al., 2003).

In this study, meteorological conditions were measured on the study sites and population dynamics of tree swallows were monitored during two summers with different weather patterns. Average rainfall in 2003 was about ten times higher than in 2004, and a dramatic drop in average temperatures occurred during the nestling period. During both years, we investigated the relationship between exposure to water and sediments affected by oil sands processing materials (OSPM) and reproductive success of adult tree swallows, as well as growth and survival of nestlings. Since organic pollutants from other industry sources are known to induce hepatic detoxification enzymes, we also measured ethoxyresorufin-*o*-deethylase (EROD) activity in nestlings as an indicator of exposure to chemicals that could be present in the OSPM.

2. Material and methods

2.1. Study sites

The study was conducted on three effluent-fed wetlands located on Suncor's and Syncrude's leases (57° 00′ Northing, 111° 30 Easting), from late May to mid July of 2003 and 2004. Wetlands of different ages (year of construction) were chosen to bracket various degrees of contamination. Naphthenic acids (NAs) concentrations in water were analyzed at Syncrude's Edmonton Research facility (Golder Associates, 2003). Smits et al. (2000) measured PAHs in sediments of various reclaimed wetlands in 1998. Since more recent data were unavailable, we summarized their results for comparative purposes.

2.1.1. "Consolidated Tailings" (CT, Suncor)

CT wetlands are adjacent to the west side of a 300 ha tailings holding pond (Pond 2/3) (Fig. 1). These wetlands were created in 1999 by flooding a 52 ha area with consolidated tailings, resulting in a mosaic of terrestrial and wetland habitats. This site represents the first stages of reclamation: fresh tailings are pumped into the wetlands from "Pond 2/3".

NAs levels are the highest at this site, reaching 68.0 mg/L. NAs concentrations in fresh tailings typically range from 90 to 110 mg/L (Rogers, 2003).

2.1.2. "Natural Wetlands" (NW, Suncor)

NW are adjacent to the south side of Pond 2/3 (Fig. 1). This 1.3 ha depression filled with uncontrolled dyke seepage and surface run-off in 1984. Additional consolidated tailings water was pumped into the wetland until September 2001. This scenario represents the second stages of reclamation: tailings water has started to mature in wetland systems, where microbial degradation is occurring. NAs concentrations average 51.9 mg/L. Total parent PAHs levels were 207.7 ng/g and total alkylatated PAHs reached 2.43 mg/g in 1998.

2.1.3. "Demonstration Pond" (DP, Syncrude)

DP is a 4–5 ha pond constructed in 1993 (Fig. 1). It consists of a 9 m deep layer of fine tailings overlain by 2.5 m of diverted local surface stream flow. This is the oldest reclaimed wetland and NAs (10.3 mg/L), parent PAHs (140.1 ng/g) and alkylated PAHs (1.09 mg/g) have decreased to lower levels, approaching those of the control site.

2.1.4. "Poplar Creek" (PC, Reference Site)

PC is a 2.5 km × 0.5 km reservoir located approximately 10 km south of mined areas (Fig. 1). It was created in 1975 when the water flowing through the leases had to be diverted prior to mining operations. It does not receive any mining effluents and is not affected by mining activities. Levels of NAs (0.3 mg/L), parent PAHs (81.5 ng/g) and alkylated PAHs (0.49 mg/g) are typical of natural lakes from the Athabasca basin.

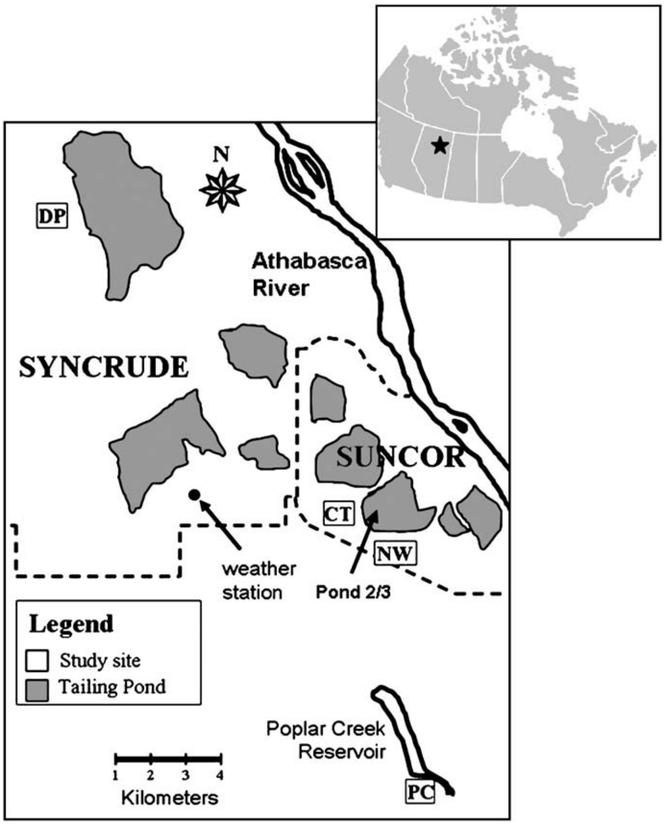

Fig. 1. Map of the mining areas, showing the location of the study sites, the meteorological station and the main tailing ponds. The dashed lines indicate the limits of Suncor's and Syncrude's company leases. On the upper right corner is a map of Canada, where the location of the Athabasca Oil Sands is indicated.

2.2. Meteorological data

Meteorological data were recorded at Syncrude's meteorological station, which has a central location relative to the study sites (Fig. 1). Data presented here are the average temperatures, mean daily precipitation and cumulative rainfall. Since we were interested in the effect of weather on nestling survival, we used data from June 18th, when the first egg hatched, to July 11th, when the last nestling would have fledged.

2.3. Tree swallows

A total of 92 nest boxes were monitored on all study sites, with 26 boxes on Poplar Creek reference site (PC), 20 on Demonstration Pond (DP), 25 on Consolidated Tailings (CT) and 21 on Natural Wetlands (NW). Tree swallow activity was monitored from May 25th to July 15th. Nest boxes were inspected daily until clutch completion to determine clutch initiation and clutch weight. Since females occasionally interrupted laying for one day or two, a clutch was considered complete when no new egg was laid after three days. Clutch weight was measured on that day, i.e. the third day with the same number of eggs in the nest. Females were aged according to plumage characteristics (Hussell, 1983). Nests were left undisturbed during the 14 day incubation period and then monitored daily to determine hatch date and hatching success (number of eggs hatched/number of eggs laid). Nestling weight and wing length were measured on days 6 and 12 (hatch day = day 1). Egg and nestling mass were measured to the nearest 0.05 g using a Pesola spring balance, wing length was measured to the nearest 0.1 cm. The number of nestlings that survived to day 12 was used to calculate fledging success (number of fledglings/eggs hatched) and nest success (number of fledglings/eggs laid). Nestlings lost to predators were excluded from the analysis of reproductive performance.

2.4. EROD

In 2004, one to three 14-day-old nestlings per nest were randomly collected to measure EROD activity. Nestlings were anesthetized with inhalation of halothane (Halocarbon Laboratories, River Edge, NJ, USA) and euthanized by cervical dislocation. Liver was removed and the left lobe (0.2–0.3 g) was frozen in liquid nitrogen within 10 minutes of collection.

Thawed liver samples were homogenized with 1 ml of KCL (0.15M) – HEPES (0.02M) buffer. Microsomes were separated from other cellular components after selective ultracentrigugation. The microsomal pellet was resuspended in 0.6 ml of tris-EDTA buffer containing 20% glycerol and samples were stored at −70 °C until further use. Enzymatic activity was quantified using an adaptation of the Canadian Wildlife Service's protocol (Trudeau and Maisonneuve, 2001).

Microsomes samples were placed in 96-well plates kept on ice (NUNC-immuno plates, Nalge Nunc International, Rochester, NY, USA). Total reaction volume in the wells was 140 μL. Sample wells contained 75–200 μg/ml of microsomal proteins, 1 μM ethoxyresorufin (Molecular Probes R-352, Invitrogen Canada, Inc., Burlington, ON, Canada) and 0.5 mM NADPH (Sigma-Aldrich Canada Ltd. N-6505, Oakville, ON, Canada). The reaction was allowed to proceed for 10 minutes at 39 °C, then was stopped by adding 60 μL of solution containing 600 μg/ml of fluorescamine (acetonitrile: AXO145, BDH Inc. Toronto, ON, Canada, fluorescamine: Sigma-Aldrich F-9015). Fluorescence of resorufin product was measured using 390 nm excitation/460 nm emission filters while fluorescamine, indicating the amount of protein, was measured using 530 nm excitation/590 nm emission filters (96-wells fluorometer, MFX Microtiter Plate reader, Dynex Technologies Inc., Chantilly, VA, USA). EROD activity was expressed as pmol/min/mg microsomal protein.

2.5. Statistical analysis

We used individual nestlings as experimental units and avoided pseudoreplication by using techniques appropriate for clustered data, as recommended by Dohoo et al. (2003). The associations between site (DP, PC, NW, CT) and continuous outcomes (e.g. weight, wing length) were examined using linear mixed models, accounting for the lack of independence among chicks from the same nest (PROC MIXED, SAS for Windows version 8.02, SAS Institute Inc., Cary, NC.). The associations between site and discrete outcomes were similarly analyzed using generalized estimating equations. A Poisson distribution with log link function was used for modeling count outcomes (e.g.: clutch size, brood size, fledglings/nest). A binomial distribution with logit link function was used for dichotomous data such as reproductive success (PROC GENMOD, SAS for Windows version 8.02). The effects of site on nestling mortality were analyzed using logistic regression models, where each nestling was assigned a binary response variable (dead = 0, survived = 1). The odds ratios obtained from logistic regression analysis (natural logarithm of the slope) were used as estimates of the relative risk of mortality.

Few breeding pairs nested on CT and NW in 2003. Sample size was too small to carry out statistical tests on fledging success, nest success and number of fledglings/nest on CT wetlands, so we combined CT and NW data where necessary (which we identify as "Suncor"). However, the NAs and PAHs levels were slightly different between these two wetlands and this must be considered when interpreting the combined results.

Other risk factors for reproductive success and nestling growth include female age, brood size and hatch date. Clutches laid later in the season or reared by younger females tend to be less successful, and chicks from larger broods may have slightly reduced growth rates (Robertson and Rendell, 2001). Female age, hatch date and brood size were therefore considered as potential confounders and included in the models when $P < 0.05$ or inclusion of the factor in the model changed the effect estimate by more than 10%.

3. Results

3.1. Tree swallow reproduction

A total of 53 and 50 breeding pairs nested on the study sites in 2003 and 2004, respectively. Female age and brood size were not associated with reproductive success in either year. In 2003, the date of clutch initiation (in julian days) was negatively associated with most parameters of reproductive success. Clutch size ($P = 0.008$), brood size ($P = 0.005$) and numbers of fledglings ($P = 0.04$), as well hatching success ($P = 0.02$) and nest success ($P = 0.03$) all decreased as the season progressed. However, fledging success was not associated with clutch initiation date ($P = 0.3$). In 2004, clutch initiation date did not significantly affect any reproductive parameter ($P > 0.05$).

Clutch size and brood size did not differ between sites in 2003 ($P = 0.6$ and $P = 0.3$, respectively) (Table 1). However, very few chicks survived to 12 days on reclaimed wetlands. Fewer chicks fledged from nests on NW ($P = 0.006$) than on PC. CT wetlands were excluded from individual site comparisons due to an insufficient number of active nests, but fewer chicks fledged from nests on Suncor when the combined results from CT and NW were compared to PC ($P = 0.006$). There were no differences in the number of young fledged between DP and PC ($P = 0.09$). Hatching success (chicks hatched/eggs laid) was similar across sites ($P = 0.4$) (Table 3). Compared with PC, fledging success (chicks fledged/chicks hatched) and nest success (chicks fledged/eggs laid) were lower on NW than on PC ($P = 0.02$ and $P = 0.007$, respectively). Both fledging success ($P = 0.003$) and nest success ($P = 0.01$) were also lower when data from NW and CT were combined and compared with PC. In DP nests, fledging success was similar to control nests ($P = 0.6$). Nest success was, however, significantly lower on DP than on PC sites ($P = 0.04$).

The 2004 breeding season was more prolific for tree swallows and differences between reclaimed wetlands and the control site were not as dramatic as in 2003. Clutch size was similar across sites ($P = 0.4$), but brood size was smaller on DP than on PC ($P = 0.01$), and fewer chicks reached fledging age on DP than on PC ($P = 0.02$) (Table 1). There were no differences in hatching success ($P = 0.1$), fledging success ($P = 0.5$) or in nest success ($P = 0.07$) (Table 2).

3.2. Nestling growth

In 2003, hatching date (in julian days) was negatively associated with nestling size. As the season progressed, 6 day-old nestlings weighed less ($P = 0.02$) and had shorter wing length ($P = 0.007$). As the season progressed, 6 day-old nestlings weighed less ($P = 0.02$) and had shorter wing length ($P = 0.003$). The size of twelve-day-old chicks was associated with hatch date (weight d12: $P = 0.7$, wing d12: $P = 0.08$). In 2004, hatch date did not influence any of the growth parameters at d6 or d12. Neither female age nor brood size affected nestling growth in any year of the study.

In 2003, weight ($P = 0.2$) and wing length ($P = 0.2$) of 6 day-old nestlings were not significantly different among sites (Tables 3 and 4). Although average measurements suggested smaller sizes on reclaimed wetlands, nestlings tended to hatch later on these sites, and differences were no longer significant after correction to a common hatch date. No d12 growth measurements could be carried out on CT because all nestlings died before that age. There was no significant difference in weight and wing length between twelve-day-old chicks from NW and DP and PC (weight: $P = 0.9$ and wing: $P = 1.0$).

In 2004, there was no significant difference in the size of six-day-old nestlings between reclaimed wetlands and the reference site (weight: $P = 0.8$; wing: $P = 0.3$) (Tables 3 and 4). Wing length for chicks from reclaimed wetlands was similar to controls at d12 ($P = 0.4$). However, d12 nestlings from reclaimed wetlands were significantly lighter compared to controls. Chicks on CT ($P = 0.03$) weighed 1.1 g less than those on PC, and those on NW ($P = 0.03$) and DP ($P = 0.01$) weighted 1.4 g less than nestlings on PC.

3.3. Nestling mortality

Low reproductive success in 2003 was related to an episode of widespread mortality among nestlings. The die-off occurred between June 22nd and July 2nd, during a period of harsh

Table 1
Reproductive performance of tree swallows (mean ± SD)

	Clutch size		Brood size		Number of fledglings/nest	
	2003	2004	2003	2004	2003	2004
Poplar Creek (PC)[a]	6.3 ± 0.8 a $n = 26$	6.7 ± 0.7 a $n = 22$	5.9 ± 0.9 a $n = 26$	6.3 ± 1.1 A $n = 22$	3.2 ± 2.6 A $n = 26$	6.3 ± 1.1 A $n = 16$
Demo Pond (DP)	6.0 ± 1.0 a $n = 12$	6.4 ± 0.8 A $n = 14$	4.3 ± 2.5 a $n = 12$	4.3 ± 2.5 B $n = 14$	1.8 ± 2.7 A $n = 12$	4.3 ± 2.5 B $n = 12$
Natural Wetlands (NW)	5.3 ± 0.7 a $n = 10$	6.2 ± 0.8 a $n = 5$	3.6 ± 2.0 a $n = 10$	5.6 ± 1.1 A $n = 5$	0.4 ± 1.1 B $n = 7$	5.6 ± 1.1 A $n = 5$
Consolidated Tailings Wetlands (CT)	4.8 ± 1.9 a $n = 5$	6.3 ± 0.7 a $n = 9$	3.2 ± 2.9 a $n = 5$	6.1 ± 0.8 A $n = 9$	0 ± 0 * $n = 3$	6.0 ± 0.8 A $n = 8$

*Sample size too small for comparison. n = number of nests. Capitalized superscripts indicate a statistical difference among sites (different letters: $P < 0.05$).
[a] Reference site.

Table 2
Reproductive success of tree swallows (mean ± SD)

	Hatching success (mean %)		Fledging success (mean %)		Nest success (mean %)	
	2003	2004	2003	2004	2003	2004
Poplar Creek (PC)[a]	92.8 ± 8.8 a $n = 26$	94.2 ± 10.8 a $n = 22$	55.2 ± 42.5 A $n = 26$	96.8 ± 6.9 a $n = 16$	51.1 ± 39.6 A $n = 26$	92.3 ± 12.9 a $n = 16$
Demo Pond (DP)	70.5 ± 39.9 a $n = 12$	69.5 ± 39.6 a $n = 14$	44.0 ± 49.7 A $n = 10$	100.0 ± 0 a $n = 9$	25.9 ± 11.0 B $n = 12$	64.4 ± 40.7 a $n = 12$
Natural Wetlands (NW)	67.0 ± 37.0 a $n = 10$	89.8 ± 9.5 a $n = 5$	12.0 ± 26.8 B $n = 5$	93.3 ± 14.9 a $n = 5$	7.1 ± 18.9 C $n = 7$	83.1 ± 11.7 a $n = 5$
Consolidated Tailings Wetlands (CT)	53.8 ± 49.5 a $n = 5$	96.8 ± 9.5 a $n = 8$	0 * $n = 1$	100.0 ± 0 a $n = 8$	0 ± 0 * $n = 3$	96.4 ± 10.1 a $n = 8$

Hatching success = eggs hatched/eggs laid per nest. Fledging success = chicks fledged/eggs hatched in nests where ≥1 egg hatched. Nest success = chicks fledged/eggs laid per nest. *Sample size too small for comparison. n = number of nests. Capitalized superscripts indicate a statistical difference among sites (different letters: $P < 0.05$). Sample size may vary for a given site because nests attacked by predators where excluded from the calculations of mean fledging and nest success. Nests with clutch failures (where no egg hatched) were not used for the calculation of fledgling success (denominators cannot be 0) but were included in the calculation of nest success.
[a] Reference site.

weather. Within in a week, a total of 134 nestlings were found dead in their nest during routine monitoring. Gross and histopathological examination was performed on 40 freshly dead nestlings. There were no lesions other than a very mild lymphoid depletion of the bursa of Fabricius in eight nestlings (20%), which was likely stress related.

Mortality occurred on all study sites, but the death toll was particularly high on the NW (89.3%) and CT (100%) wetlands. Table 5 shows descriptive mortality rates for each site, which were calculated as the total number nestlings found dead on a site, divided by the number of nestlings alive on that site prior to the harsh weather. The odds ratios presented in Table 6 are an estimate of the relative risk of dying on reclaimed wetlands compared to the risk of dying on the reference site (please refer to the methods section). Nestlings on NW were 11.4 times more likely to die than those on the PC site (95% CI of the odds ratio = 1.4–91.9, $P = 0.02$). Using combined CT and NW data produced similar conclusions. Chicks from Suncor (CT and NW) were 13.3 more likely to die than those from the PC site (95% CI = 1.6–107, $P = 0.02$). Only four nestlings died in 2004. There was no association between site and nestling fate ($P = 0.5$).

3.4. Meteorological conditions

The 2003 nestling die-off was triggered by an abrupt temperature plunge of 15.8 °C (from 23.7 °C on June 18th to 7.9 °C July 22nd) synchronized with heavy and persistent rain (Fig. 2a). In 2004, temperatures oscillated up and down, but there was no sudden and sharp decrease (Fig. 2b). Although the patterns of temperature variation changed between 2003 and 2004, average temperatures over the breeding

Table 3
Mass of nestling tree swallows (mean ± SD)

	Year	Poplar Creek (PC)[a]	Demo Pond (DP)	Natural Wetlands (NW)	Consolidated Tailings Wetlands (CT)
Weight Day 6 (g)	2003	13.75 ± 3.18 a $n = 102$	11.64 ± 3.20 a $n = 48$	8.73 ± 2.17 a $n = 12$	9.55 ± 6.42 a $n = 11$
	2004	13.29 ± 2.01 a $n = 93$	12.98 ± 2.08 a $n = 60$	12.34 ± 2.20 a $n = 26$	12.84 ± 2.75 a $n = 55$
Weight D12 (g)	2003	23.61 ± 2.37 a $n = 40$	22.00 ± 2.23 a $n = 21$	21.08 ± 1.53 a $n = 3$	–
	2004	24.08 ± 1.56 A $n = 84$	22.68 ± 1.61 B $n = 53$	22.66 ± 2.42 B $n = 26$	22.95 ± 1.73 B $n = 55$

n = number of nestlings. Capitalized superscripts indicate a statistical difference among sites (different letters: $P < 0.05$).
[a] Reference site.

Table 4
Wing length of nestling tree swallows (mean ± SD)

	Year	Poplar Creek (PC)[a]	Demo Pond (DP)	Natural Wetlands (NW)	Consolidated Tailings Wetlands (CT)
Wing Day 6 (cm)	2003	2.1 ± 0.5 a $n = 102$	1.8 ± 0.4 a $n = 48$	1.5 ± 0.3 a $n = 12$	1.6 ± 0.8 a $n = 11$
	2004	1.9 ± 0.3 a $n = 93$	1.9 ± 0.3 a $n = 60$	1.7 ± 0.2 a $n = 26$	1.8 ± 0.3 a $n = 55$
Wing D12 (cm)	2003	4.8 ± 0.8 a $n = 41$	4.5 ± 0.6 a $n = 21$	4.6 ± 0.4 a $n = 3$	–
	2004	5.0 ± 0.4 a $n = 84$	4.9 ± 0.3 a $n = 53$	4.8 ± 0.5 a $n = 26$	4.8 ± 0.5 a $n = 55$

n = number of nestlings. Capitalized superscripts indicate a statistical difference among sites (different letters: $P < 0.05$).
[a] Reference site.

Table 5
Mortality rate of nestling tree swallows

	Mortality rate[a] (%)	
	2003	2004
Poplar Creek (PC)[b]	48.0%	2.8%
	73/152	3/106
Demo Pond (DP)	58.8%	0%
	30/51	0/52
Natural Wetlands (NW)	89.3%	3.6%
	25/28	1/28
Consolidated Tailings Wetlands (CT)	100%	0%
	6/6	0/54

[a] Mortality rate = total number of dead chicks/total number of chicks alive prior to the die-off (nestlings victims of predators excluded).
[b] Reference site.

season were not significantly different ($P = 0.3$). However, there were striking differences in rainfall (Table 7). Daily rainfall was about ten times higher in 2003 than in 2004 ($P = 0.01$).

3.5. EROD

Nestlings from CT had the highest ethoxyresorufin-*o*-deethylase activity, closely followed by those from NW (Table 8). In chicks from CT, EROD was 2 times higher than in those from the reference site ($P = 0.02$), and in those from NW, it was 1.9 times higher than in nestlings from PC. Differences between NW and PC approached statistical significance ($P = 0.07$). EROD activity in nestlings from DP was similar to that in nestlings from the PC site ($P = 0.8$).

4. Discussion

4.1. Tree swallow reproduction

Reproductive success was very low on reclaimed wetlands in 2003, especially on CT and NW. To the authors' knowledge, we documented one of the lowest fledging success rates reported in a passerine species exposed to contaminants of industrial origin in the wild. In most studies, the impact of contaminants on reproduction has not been as dramatic. Fledging success in tree swallows exposed to PCBs ranged from 51% to 74% (Bishop et al., 1999) and nest success varied between 50% and 80% (McCarty and Secord, 1999). Nest success ranged from 75% to 77% in great tits (*Parus major*) breeding in the vicinity of a smelter (Janssens et al., 2003). Reclaimed wetlands may contribute to low recruitment into the tree swallow population if reproductive performance were consistently as impaired as in 2003. Nevertheless, since this poor reproductive success was associated with sustained inclement meteorological conditions (refer to the mortality section), which is relatively infrequent, it is unlikely that such low tree swallow productivity would occur every year.

In contrast, reproductive success was not different between contaminated wetlands and the reference site in 2004, and it was comparable to what has been reported for tree swallow populations on non-contaminated sites (Robertson et al., 1992). Broods were significantly smaller on DP than on PC in 2004, but this was linked to three complete nest failures (nests where no eggs hatched), which could not be clearly related with contaminant exposure. We frequently observed American kestrels (*Falco sparverius*) hunting on DP, trying to reach inside the nest boxes. It is likely that kestrels caught incubating females or disturbed them enough to cause nest abandonment.

Table 6
Effect of site on nestling mortality[a]

	Slope	95% CI	P	Odds ratio[b]	95% CI of odds ratio
Poplar Creek (PC)[c]	0	0	–	–	–
Demo Pond (DP)	0.4	−1.0, 1.8	0.57	1.5	0.4, 5.8
Natural Wetlands (NW)	2.4	0.4, 4.5	0.02	11.4	1.4, 91.9
Consolidated Tailings Wetlands (CT)	*	*	*	*	*
Suncor (CT & NW)	2.6	0.5, 4.7	0.02	13.3	1.6, 107

*Sample size too small for comparison.
[a] Coefficients from multiple logistic regression with PC as the control site.
[b] Odds ratio = the natural logarithm of the slope.
[c] Reference site.

4.2. Nestling growth

There were no differences in nestling size among sites in 2003. It is possible that the small sample size may have masked any differences in nestling size. Only three nestlings (from one nest) were measured on NW, and none on CT. Moreover, mortality may have masked differences in growth rates. The smallest nestlings of a brood are the most likely to die during episodes of food shortage (Bryant, 1978). Since survival rates were very low on NW, the smallest nestlings would have died, skewing the mean toward higher weights. Differences in growth rates between reclaimed wetlands and the control site might have been detected if mortality rates had been similar.

In 2004, chicks on reclaimed wetlands weighted 4 to 5% less than those on the reference site. Growth rates of nestlings have been unaffected by contaminant exposure in other studies with tree swallows (Bishop et al., 2000b; Harris and Elliott, 2000; McCarty and Secord, 1999). Food availability is one of the main factors determining chick's growth (McCarty and Winkler, 1999), and it was hypothesized that nestlings grew normally because insect abundance was not affected by contamination. The situation might be different on the oil sands, because growth and survival have been significantly reduced in benthic invertebrates exposed to oil sands process water (Whelly, 1999). CT is a new experimental site, and data on benthic invertebrate populations in these wetlands are not available yet. DP was unable to support a viable population of aquatic invertebrates in 1997–1998 (Whelly et al., 1998), and chicks could have been smaller on this site because of reduced food availability. Conversely, benthic invertebrate biomass is higher in NW than in neighboring references wetlands (Bendell-Young et al., 2000). This suggests that other factors, including contaminants such as NAs, may be responsible for sustained compromised tree swallow growth

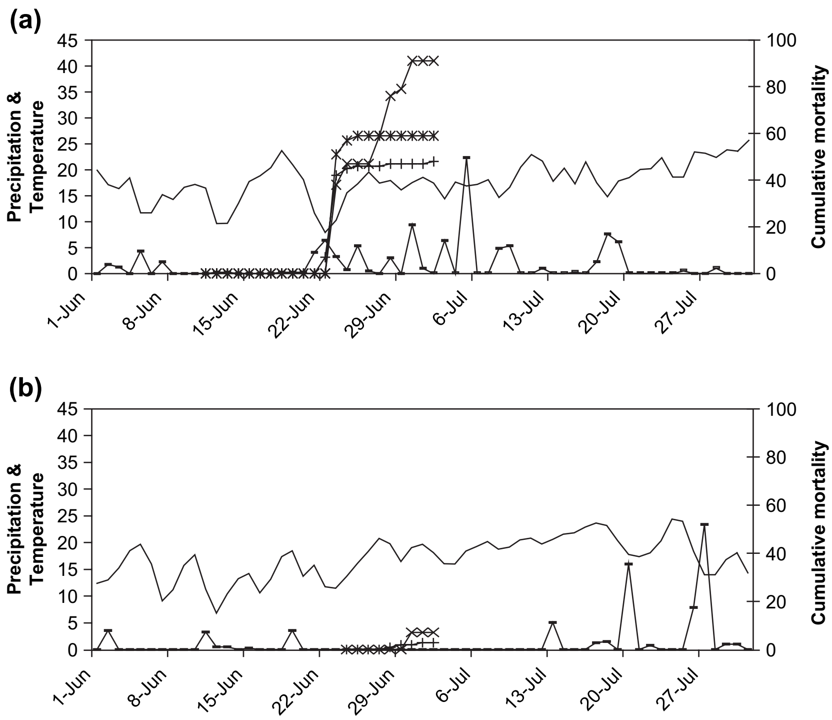

Fig. 2. Cumulative mortality over time of nestling tree swallows. (a) During an episode of harsh weather in 2003. (b) During normal weather in 2004. The left axis shows the temperature in degrees Celcius (°C) and the average precipitation in millimeters (mm). On the graph, lines without markers represent the temperature, while lines with "–" markers represent precipitation. The right axis shows cumulative mortality (%). Lines with "×" markers represent cumulative mortality on Natural Wetlands (NW) and Consolidated Tailings wetlands (CT) (Suncor). Lines with "*" markers represent cumulative mortality on Demo Pond (DP, Syncrude). Lines with "+" markers represent cumulative mortality on Poplar creek (PC, reference site). The bottom axis shows the date.

rates on NW, which were first reported in 1998 (Smits et al., 2000). Body weight was significantly lower in rats orally exposed to NAs extracted from tailings compared with controls (Rogers, 2003).

Regardless of the cause, nestlings from reclaimed wetlands may suffer lower post-fledging survival as a result of their smaller size. It has been repeatedly shown that post-fledging survival is strongly correlated with body mass at fledging (Lindèn et al., 1992; Naef-Daenzer et al., 2001; Ringsby et al., 1998; Tinbergen and Boerlijst, 1990). Since Smits et al. (2000) also found nestlings from the reclaimed wetlands to be smaller than control birds, the results of this long-term monitoring suggest a well-established trend.

Table 7
Meteorological data (mean ± SD) for the chick rearing period, June 6th to July 11th

	2003	2004
Mean daily temperature (°C)	16.4 ± 3.6 a	15.8 ± 3.7 a
Min-Max daily temperature (°C)	7.9–23.7	6.8–20.8
Mean daily precipitation (mm)	2.12 ± 4.25 A	0.23 ± 0.80 B
Total precipitations (mm)	74.0	8.1
Number of days with precipitations	19	5

Capitalized superscripts indicate a statistical difference among years (different letters: $P < 0.05$).

4.3. Nestling mortality and meteorological conditions

Episodes of inclement weather can result in dramatic bird mortality. Aerial insectivores such as swallows are particularly susceptible to abrupt variations in environmental conditions, as rain, wind and cold temperatures may ground the insects and make them unavailable for foraging (McCarty and Winkler, 1999). Moreover, cold temperatures exert negative effects directly on nestlings. Young altricial chicks are ectothermic and will not survive low ambient temperatures unless parents provide sustained brooding. Older chicks are homeothermic, but the energetic costs of thermoregulation may surpass energetic reserves, especially when foraging is compromised, and cause great metabolic stress (Visser, 1998). The limiting effects of cold climate on nestling development and survival might be particularly important in this study.

Table 8
Hepatic EROD activity (mean ± SD) in nestlings in 2004

	Poplar Creek (PC)[a]	Demo Pond (DP)	Natural Wetlands (NW)	Consolidated Tailings Wetlands (CT)
EROD	38.0 ± 23.2 A	45.7 ± 36.8 AB	72.3 ± 27.7 AB	74.2 ± 44.3 B
n	16	16	16	20

n = number of nestlings. Capitalized superscript letters indicate a significant difference among sites (different letters: $P < 0.05$).
[a] Reference site.

Indeed, unless most previous research, which was conducted in northern United-States or southern Canada, our study sites were located close to the northern breeding limit for tree swallows.

Nevertheless, mortality rates on PC (48%) and DP (58%) were comparable to those reported for a similarly weather-related die-off of purple martins in Pennsylvania, where 58% of the nestlings died (Hill and Chambers, 1992). Mortality rates on CT (100%) and NW (89%) are extreme compared to what has been documented for birds unchallenged by contaminants. Although large numbers of adult occasionally perish following spring cold spells at their arrival on breeding grounds (Brown and Bomberger Brown, 1998; duBowy and Moore, 1985), such a widespread mortality of nestlings has never been documented in tree swallows, even in those exposed to high levels of contaminants from point source pollution.

Localized storms or precipitation events could have hit Suncor sites more heavily than the others, resulting in higher mortality rates. However, this would be unlikely, because all the study sites are within a 12 km radius from Syncrude weather station. There were no significant differences in rainfall and temperature between Suncor's and Syncrude meteorological stations (data not shown). However, meteorological data for the control site were unavailable. It would have been necessary to record data at each of the sites to completely discount this hypothesis.

Nonetheless, site-specific environmental factors, such as the exposure of the nest boxes to wind and rain, could have affected nestling survival. Paradoxically, boxes on the reference site (PC) were the least protected from the elements, whereas those on Suncor were the most sheltered. Although PC is bordered by mixed forest composed of white spruce (*Picea glauca*), balsam poplar (*Populus balsamifera*) and trembling aspen (*Populus tremuloides*), nest boxes are perched on the exposed shore of the 1.25 km^2 reservoir, facing the water. During episodes of inclement weather, strong winds over the lake pushed the rain into entrance holes, which often dampened the feathers lining the nests and the chicks. A similar situation was found on Syncrude, where boxes faced the 5 ha pond and were surrounded by grassy areas devoid of shrubs or trees. These nestlings appeared less protected from cold ambient temperature. In contrast, dense shrub communities (*Salix sp., Populus sp., Potentilla fruticosa*) grow around boxes on NW. CT site offers intermediate sheltering, as boxes are partly protected by young (1.7 to 2 m high) white spruces. If chick mortality were correlated with nest exposure, we would have expected it to be the worst on PC and DP, not on CT and NW. This also implies that if the nests on PC were better protected, the differences in survival between this site and Suncor reclaimed wetlands would have been even more dramatic.

Concentrations of NAs and petroleum hydrocarbons are highest on CT, closely followed by NW. Exposure to these contaminants could have impaired nestlings' ability to tolerate additional natural stressors, including hypothermia and food shortage. Alternatively, the large energetic demands for thermoregulation could have triggered a decrease in hepatic detoxification, leading to intoxication. Either case could explain why we did not observe any die-off in 2004, when the weather was overall less challenging. A small number of studies have shown that chemical toxicity can be exacerbated by other stressors. Reduced food intake increased DDE toxicity in ringed doves (Keith and Mitchell, 1993). The synergistic effects of mercury, adverse weather, and food shortage have been reported as a cause of mortality in common loons (Daoust et al., 1998). Fish experimentally exposed to cold water were more susceptible to selenium toxicity (Lemly, 1993). Fish are poikilotherms, so their response to temperature stress might be different from that of an adult bird or an older nestling. However, it could possibly be comparable with that of a young, ectothermic nestling.

The current findings raise a potential concern regarding long-term survival of tree swallows fledged at OSPM sites because altricial birds cope with important challenges shortly after fledging, as they must learn to become nutritionally independent, start to disperse and undertake the winter migration (Kershner et al., 2004). Juveniles that are less fit, such as those that were reared on OSPM sites, may suffer excessive juvenile mortality after leaving their nest. Yet, the period immediately following fledging is rarely monitored in ecotoxicology studies because of logistic difficulties. As Keedwell (2003) pointed out, leaving the nest does not necessarily equal success.

4.4. EROD

Cytochrome P450 activity increased proportionally with the degree of wetland contamination, which was presumably linked to the presence of PAHs and NAs in nestlings' diet. Custer et al. (2001) and Bishop et al. (1999) similarly documented an EROD induction in tree swallows exposed to PAHs. Oral exposure to NAs also stimulated EROD in laboratory rats (Rogers, 2003). These findings indicate that birds on reclaimed wetlands have to allocate part of their metabolic resources to detoxification efforts.

Differences between the EROD activity in birds from NW and PC were close enough to statistical significance for their biological importance to be noted. They suggest that insects developing in contact with mining effluents are still relatively toxic to birds even after tailings have started to maturate in wetlands systems. Previous research showed that the toxicity of tailings water transferred to shallow pits decreases over a one to two years period (MacKinnon and Boerger, 1986). Mining effluents have not been pumped into NW for three years. Still, monooxygenase activity in nestlings was as stimulated as when fresh tailings were being pumped into these wetlands: Smits et al. (2000) documented a similar increase in EROD activity in nestlings on NW in 1998. On the other hand, nestlings on DP, which was constructed in 1993, seemed minimally affected. These findings indicate that it might take several years before contaminant levels in reclaimed wetlands do not pose a problem for species such as tree swallows.

5. Conclusions

During episodes of cold and rainy weather, tree swallow reproductive performance was greatly reduced on reclaimed wetlands containing Oil Sands Process Material compared to control wetlands, and nestlings on reclaimed wetlands had much higher mortality rates than those on the reference site. In the absence of challenging environmental conditions, nestlings from reclaimed wetlands survived to fledging, but were not as large as those on the control site and showed increased hepatic detoxification activity. These results suggest that the implementation of current wet landscape reclamation to a large geographic scale might negatively impact reproductive success of tree swallows and potentially other birds nesting there, but it is difficult to determine to what extent, because the study sites where the tree swallows were the most affected represented early reclamation scenarios, where maturation of tailings in wetlands systems was not complete yet. This study also demonstrates that unless nestlings were subjected to a combination of stressors, they were relatively resilient to chemical exposure. Limiting biomonitoring to the pre-fledgling period might underestimate the consequences of exposure to contaminants, because nestlings are relatively unchallenged until they leave their nest.

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